

BATTLE CREEK KELLOGG, MI, USA

WMO: 725396

Lat: 42.308N

Lon: 85.251W

Elev: 928

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.1	(c) 6.6	(d) -6.1	(e) 4.1	(f) 3.3	(g) -0.9	(h) 5.4	(i) 8.1	(j) 26.0	(k) 22.5	(l) 24.1	(m) 25.3	(n) 7.5	(o) 210	(p) 0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	89.6	73.2	86.6	71.6	83.9	70.3	76.0	84.9	74.5	82.7	73.0	80.7	11.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	127.5	80.8	72.0	122.4	79.5	70.3	115.5	77.8	40.1	85.2	38.5	82.8	37.1	80.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.7	20.3	18.3		DB	-6.0	93.2	7.1	3.0	-11.1	95.4	-15.2	97.1	-19.2	98.8	-24.3	101.0
				WB	-6.5	79.1	6.9	1.9	-11.4	80.5	-15.5	81.6	-19.3	82.7	-24.3	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	24.2	26.3	36.3	48.7	60.0	68.8	72.6	70.8	64.1	52.0	40.4	29.8	(6)
	DBStd	19.25	10.95	10.45	11.41	9.33	8.72	6.85	5.47	5.14	7.52	8.72	9.18	10.07	(7)
	HDD50	3094	800	666	453	134	16	0	0	0	2	81	313	629	(8)
	HDD65	6348	1264	1084	893	495	206	40	6	11	106	414	739	1090	(9)
	CDD50	2950	1	2	27	93	324	563	700	644	426	142	24	4	(10)
	CDD65	731	0	0	2	5	50	153	241	190	80	10	0	0	(11)
	CDH74	5988	0	0	13	50	445	1275	2106	1389	636	74	0	0	(12)
	CDH80	1720	0	0	1	5	106	391	690	345	171	11	0	0	(13)
(14) Wind	WSAvg	8.7	10.2	9.9	9.6	10.0	8.7	7.8	7.1	6.6	7.1	8.5	9.6	9.8	(14)
(15) Precipitation	PrecAvg	35.5	2.1	1.7	2.0	3.0	3.9	3.7	3.4	3.6	3.7	3.5	2.8	2.0	(15)
	PrecMax	43.8	4.3	4.1	4.2	5.9	9.2	7.2	6.1	6.6	12.1	7.1	6.5	3.3	(16)
	PrecMin	28.3	0.7	0.4	0.5	0.5	1.4	1.6	0.6	0.7	1.5	0.7	0.7	0.6	(17)
	PrecStd	4.9	1.1	1.0	1.0	1.3	2.1	1.6	1.6	1.6	2.4	1.7	1.6	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	59.8	75.3	79.9	87.7	91.6	94.0	90.9	89.2	81.6	69.1	61.0	(19)
		MCWB	53.1	52.4	62.3	62.0	69.7	73.5	75.1	74.1	71.0	69.7	57.7	57.3	(20)
	2%	DB	48.9	52.4	66.4	73.8	82.9	88.0	90.0	86.4	85.0	75.2	62.9	54.2	(21)
		MCWB	45.1	48.1	56.9	59.4	66.8	71.6	74.4	72.8	70.8	63.9	55.1	50.7	(22)
	5%	DB	43.4	45.4	60.1	69.8	79.3	84.4	86.7	83.6	81.1	70.6	58.7	48.0	(23)
		MCWB	39.9	41.2	52.1	56.9	65.2	70.1	72.2	70.6	68.2	61.7	52.3	44.7	(24)
	10%	DB	38.6	40.1	53.9	64.7	75.0	81.4	83.7	81.3	77.2	66.3	54.7	43.2	(25)
		MCWB	36.3	36.3	47.0	53.5	63.2	68.8	70.9	69.7	66.1	59.0	48.8	39.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	55.5	63.5	66.4	72.4	76.4	78.9	77.6	75.4	71.0	60.3	57.7	(27)
		MCDB	55.5	57.8	75.7	75.3	82.8	87.5	88.7	87.1	84.9	79.7	65.2	61.3	(28)
	2%	WB	46.2	48.1	58.4	62.0	69.5	74.1	76.5	75.2	73.0	67.1	56.7	51.0	(29)
		MCDB	48.1	51.7	65.4	69.6	79.1	83.7	85.5	82.5	80.5	71.9	61.3	53.6	(30)
	5%	WB	40.4	41.8	53.7	58.6	67.5	72.2	74.9	73.6	70.7	63.3	53.2	45.1	(31)
		MCDB	43.0	45.0	58.9	67.0	76.2	81.5	83.4	80.2	77.1	68.1	57.8	47.4	(32)
	10%	WB	36.3	36.7	47.4	55.1	65.2	70.6	73.2	71.8	68.5	59.8	49.5	40.1	(33)
		MCDB	38.7	39.9	52.8	63.7	72.8	78.8	81.0	78.4	74.9	65.6	54.7	43.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.0	16.8	19.8	19.4	18.9	19.3	18.3	19.4	17.3	14.1	11.4	(35)
		MCDBR	16.1	18.7	24.0	25.1	23.1	21.6	21.8	20.7	22.8	22.6	19.1	16.5	(36)
		MCWBR	13.7	15.4	16.1	14.7	12.2	9.7	9.7	9.5	10.7	13.1	14.2	14.2	(37)
	5% WB	MCDBR	14.9	17.8	21.3	21.9	20.0	18.5	18.4	16.8	18.6	19.3	17.7	15.3	(38)
		MCWBR	13.7	15.8	15.5	15.3	11.8	9.7	9.5	9.3	10.4	13.6	14.6	14.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.295	0.358	0.348	0.390	0.424	0.443	0.430	0.438	0.389	0.350	0.317	0.303	(40)
	taud		2.369	2.146	2.340	2.250	2.197	2.187	2.231	2.203	2.319	2.427	2.467	2.397	(41)
	Ebn at Noon		268	261	284	279	271	265	267	261	266	265	259	254	(42)
	Edh at Noon		26	38	35	42	45	46	43	43	36	29	23	23	(43)
(44) All-Sky Solar Radiation	RadAvg		440	711	1117	1466	1700	1923	1952	1680	1357	864	534	361	(44)
	RadStd		42	100	113	137	165	126	89	88	124	90	75	43	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page